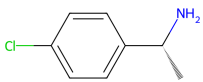
(a) Molecule and canonical SMILES



`C[C@@H](N)c1ccc(Cl)cc1`

RDKit canonical form after salt stripping and charge neutralisation (22 characters)

(b) Atom-level tokenisation (16 tokens)

C	[C@@H]	(	N	)	c	1	c	c	c	(	Cl	)	c	c	1
7	18	5	6	10	3	4	3	3	3	5	25	10	3	3	4

regular expression of Schwaller et al.; shaded tokens span several characters

0 · · · padded to $T = 198$

(c) Vocabulary lookup ($V = 75$ tokens for SIDER; PAD $= 0$); arrows map each token to its index

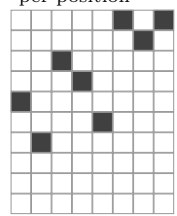
(d) Three input encodings of the same sequence (first eight positions shown)

integer index
one scalar i_t/V per position

0.09	0.23	0.06	0.08	0.13	0.04	0.05	0.04

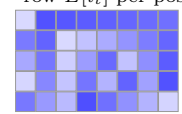
shape $T \times 1$; imposes an ordinal relation between unrelated tokens

one-hot
binary vector of length V per position



shape $T \times V$; sparse, fixed width

learned embedding
trainable table $E \in \mathbb{R}^{V \times d}$, row $E[i_t]$ per position



shape $T \times d$ ($d = 64$ default, 16–128 searched); dense, learned jointly with the network

convolution / recurrent layers → classifier